



LOW BOOM DESIGN AND ANALYSIS OF A BLENDED-WING-BODY CONFIGURATION FOR LARGE-SIZED SUPERSONIC TRANSPORT

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Abstract

For the development of next-generation supersonic transport (SST), most existing SST configurations feature small passenger capacity, which greatly limits their commercial application. For conventional tube-and-wing (TAW) configuration, fuselage volume increases quadratically with passenger capacity. By comparison, the blended-wing-body (BWB) shape fits the cabin better, resulting in far less volume growth as the passengers rises. This work aims to explore a supersonic blended-wing-body (SBWB) configuration, with the expectation to achieve higher seating capacity under the same level of sonic boom constraint of the conventional TAW configuration. A 160-seat SBWB configuration is proposed, and its low-boom design potential is analyzed under cabin volume constraints. The Jones–Seebass–George–Darden (JSGD) inverse design method is adopted to optimize the fore-body pressure waveform, resulting in a 6.46 PLdB reduction in the ground loudness of the SBWB. Subsequent waveform evaluation reveals prominent over-expansion features along the rear-body, which results in a bottleneck for additional sonic boom performance gains. Auxiliary components are widely applied in low-boom design to adjust rear body waveform. Accordingly, two typical fitting components for BWB configuration, namely winglets and V-tails, are selected to compare their waveform regulation capabilities. Results show winglets effectively suppress rear-body over-expansion and improve near-field pressure waveforms, whereas V-tails bring limited improvement. Following this round of multi-part optimization, the overall sonic boom intensity of the configuration reaches 98.7 PLdB. The proposed configuration exhibits favorable characteristics and retains considerable potential for further sonic boom mitigation.

Keywords: supersonic transport, blended-wing-body(BWB), sonic boom, aircraft design.

1. Introduction

Higher travel speed has long been a pursuit of humanity. The flight speed of supersonic transport is twice that of traditional subsonic airliners or even higher, which significantly improves travel efficiency and facilitates global economic and cultural exchanges. Therefore, the new-generation SST represents a core technological frontier in the aviation industry of the 21st century^[1-2]. In its aviation strategic plan released in 2023, NASA stated that "the development of efficient, economical and eco-friendly supersonic commercial transportation will become a gamechanger for intercontinental travel in the future"^[3]. In 2025, the White House signed a presidential executive order titled "Leading the World in Supersonic Flight" to boost the development of domestic supersonic civil air transportation. After decades of research, various advanced low-boom SST configurations have been developed across different countries, such as NASA X-59^[4], Lockheed Martin QSTA^[5], TsAGI Strizh of Russia^[6] and JAXA S4 of Japan^[7]. Nevertheless, most of these designs are intended for small-capacity supersonic business jets. The X-59, which delivers the optimal sonic boom suppression, carries merely a single passenger and serves solely as a demonstrator for sonic boom research. While small supersonic business jet concepts benefit from certain market demand and can more readily satisfy low sonic boom limits, supersonic transport imposes stringent airport requirements and consumes far more fuel than subsonic airliners, leading to exorbitant operating costs. To offset such heavy costs, airfares would surge considerably, making these aircraft inaccessible to most travelers. Expanding seating capacity represents one of the most direct approaches to improving economic efficiency. Distributing fixed operational expenditures including airport fees and fuel costs across a larger passenger pool can reduce ticket prices while increasing overall economic returns. Over the past few

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decades, research into low-boom design has matured steadily within academic circles. Geometry shaping is the most widely adopted and effective design technique^[2]. Driven by the demands of aerodynamic configuration design, a set of powerful inverse design algorithms including JSMD^[8-11], LSG^[12-14] and PNFO^[15] have been developed. These advances render the low-boom design of large-sized SST a viable prospect.

The fuselage of TAW configurations is characterized by low volumetric efficiency. When such configurations are adopted for large-sized supersonic airliners, stringent volumetric constraints considerably increase the complexity of low-boom design. Innovative configurations for supersonic civil aircraft have also been investigated in recent years, among which the bidirectional flying wing configuration^[16-17] is a representative example. However, its rotating mechanism introduces unprecedented structural challenges for the passenger cabin and does not deliver tangible improvements in commercial profitability in practice. By contrast, the blended-wing-body (BWB) configuration largely utilizes the proven component architecture of conventional aircraft^[18-19]. Beyond its superior aerodynamic performance, the non-circular central fuselage of the BWB conforms closely to cabin contours, which partially mitigates the volumetric penalties incurred by increased seating capacity and facilitates low-boom optimization.

Accordingly, a 160-PAX large SBWB configuration is proposed in this study. Its aerodynamic and sonic boom characteristics are investigated, and the JSMD inverse design method is applied to fore-body waveform design. Following an assessment of rear-body waveform features, we further explore how winglets and V-tails modulate near-field wave profiles.

2. Numerical Simulation Methods

2.1 Aerodynamic analysis

Aerodynamic and sonic boom analyses are performed using high-fidelity CFD numerical simulations. The Reynolds-Averaged Navier-Stokes (RANS) equations combined with the turbulence model constitute the governing equations. Zonal meshes composed of cylindrical near-field and conical far-field domains are adopted for high-fidelity sonic boom analysis. Grid lines of the conical mesh are arranged along the Mach angle to suppress numerical dissipation in sonic boom propagation.

The JAXA-Wing-Body (JWB) model from the 2nd AIAA Sonic Boom Prediction Workshop (SBPW) is employed for validation. Figure 1 presents the geometric configuration and sonic boom mesh topology of the JWB model, with a total mesh cell count of 17 million.

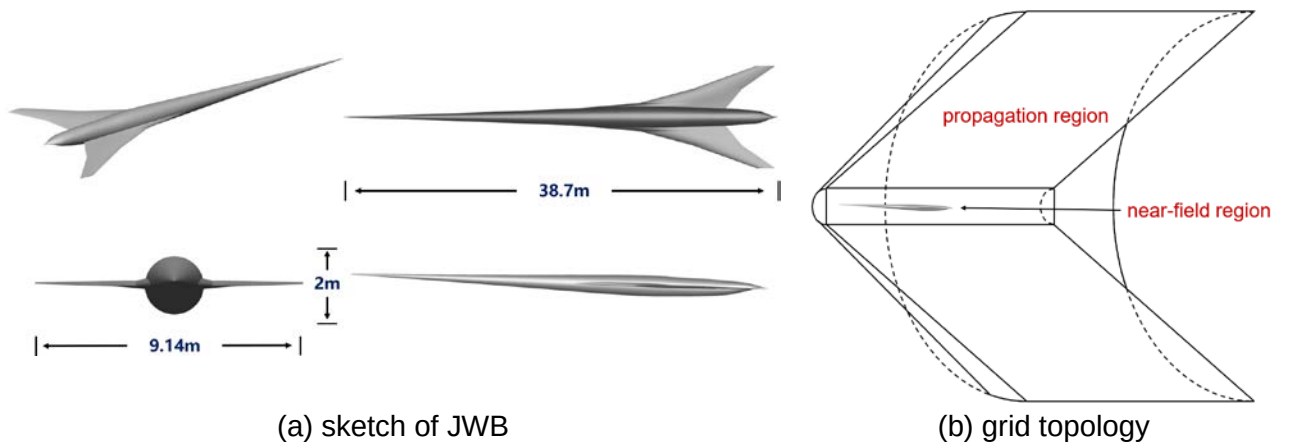


Figure 1 - Demonstration of geometry and mesh of JWB

The results are compared with the data provided by workshop participants (Figure 2). Near-field signatures are extracted directly beneath the aircraft at distances of 0.85 body length ($R/L=0.85$) and 2.55 body lengths ($R/L=2.55$). The near-field pressure waveform exhibits favorable consistency with reference results. Apparent discrepancies among the rear-body waveforms happen, which primarily originate from intricate local flow features. Even so, the overall variation trend of the obtained results aligns well with published datasets. The predictions of aerodynamic coefficients also agree well with the benchmark values.

Table 1 - Information on partial participants in SBPW-2 JWB case

	Submitter	Mesh Type	Solver	Equations	Mesh Quantity
P1	Aftosmis	Cartesian	Cart3D	Euler	8865000
P2	Magee	Unstructured	FUN3D	Euler	11335260
P3	Clemens	Unstructured	FUN3D-VL	Euler	11335260
P4	Housman	Structured	LAVA-HWCNS	RANS	17960265
P5	Jensen	Unstructured	LAVA	Euler	3953277

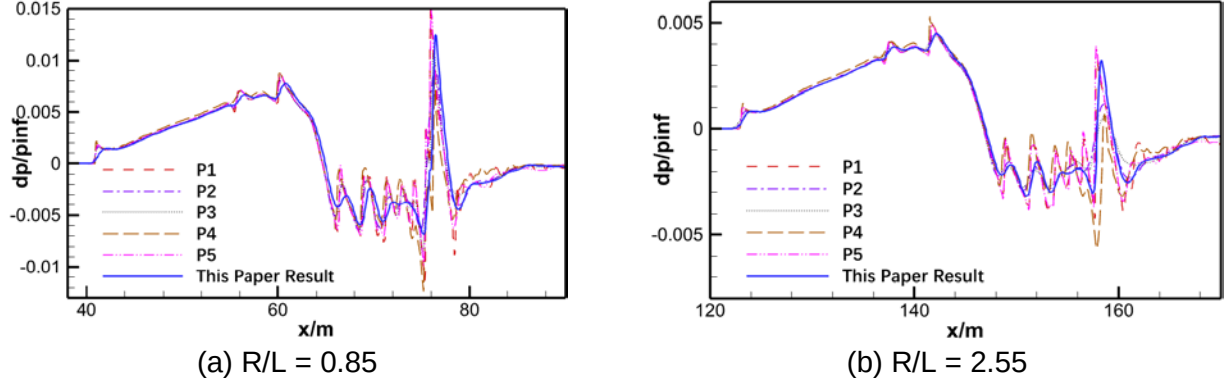


Figure 2 - Comparison of near-field waveforms of JWB

2.2 Sonic Boom Prediction

Both low-fidelity and high-fidelity methods of sonic boom prediction are applied in this study. The low-fidelity modified Whitham linear theory is used for rapid low-boom design. The far-field propagation method based on the Augmented Burgers equation is integrated with the near-field overpressure waveform obtained by CFD simulations, giving high-fidelity sonic boom results covering both near- and far-fields.

2.2.1 Rapid Sonic Boom Prediction

The JWB model is adopted to validate the rapid sonic boom prediction method derived from modified Whitham linear theory. The corresponding F-function is calculated from the equivalent-area distribution to acquire near-field sonic boom characteristics, while the lift distribution is solved using the high-order panel method. The comparisons between predicted signatures and CFD outcomes are presented in Figure 3. The waveforms computed by the rapid method based on modified linear theory consist of multi-segment polylines. As illustrated in the figure, the forebody waveforms share basically consistent trends with high-fidelity simulation results. Minor shocks originate from computational errors induced by sparse panel meshes, and errors at this level can be neglected during far-field propagation calculations. Meanwhile, considerable discrepancies are observed in the predicted aft-body wave systems. This is attributed to severe component interference in the rear-body region. Since the modified linear theory approximates the aircraft as a one-dimensional perturbation source, it fails to capture a variety of three-dimensional effects including component interference. Accordingly, the JSGD design method developed based on modified linear theory is naturally only valid for forebody wave systems. Therefore, the accuracy of the rapid computational method is acceptable during the JSGD design phase.

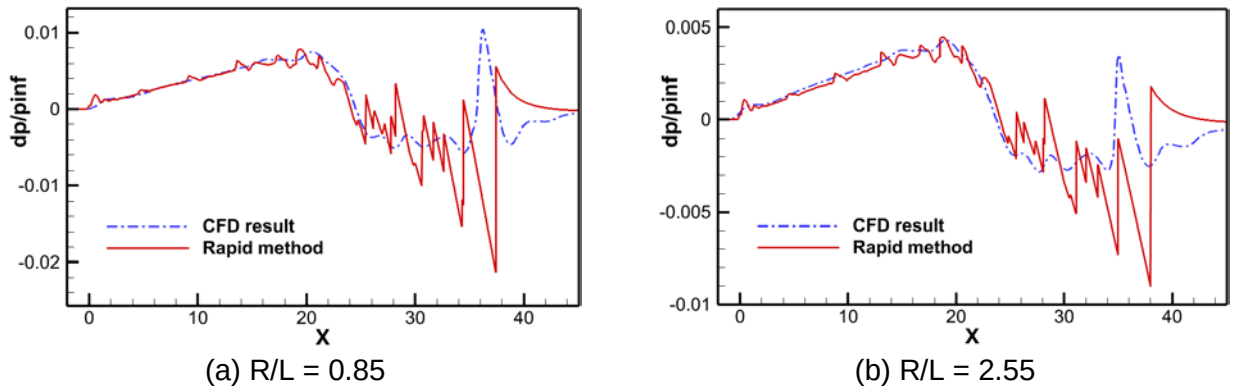


Figure 3 - Comparison between results of rapid sonic boom prediction method and CFD

2.2.2 Far-Field Sonic Boom Propagation

The far-field sonic boom propagation method based on the Augmented Burgers equation is employed in this paper. The fluid governing equations are converted into acoustic formulations via equation transformation. With third-order infinitesimal terms neglected and geometric acoustic diffusion together with atmospheric influences taken into consideration, a one-dimensional propagation equation is derived, as expressed below.

$$\frac{\partial p'}{\partial s} = -\frac{1}{2G} \frac{\partial G}{\partial s} p' + \frac{\beta p'}{\rho_0 c_0^3} \frac{\partial p'}{\partial t'} + \frac{\delta}{2c_0^3} \frac{\partial^2 p'}{\partial t'^2} + \sum_j \frac{(\Delta c)_j \tau_j}{c_0^2 (1 + \tau_j \frac{\partial}{\partial t'})} \frac{\partial^2 p'}{\partial t'^2} \quad (1)$$

where p' stands for sound pressure and s refers to acoustic ray coordinate. The term $t' = t - \int ds/c_0$ denotes the delay time of sonic boom waveform, with c_0 representing ambient speed of sound. Terms on the right side of the equation correspond to diverse atmospheric impacts on sonic boom propagation, covering geometric acoustics, atmospheric stratification, thermoviscous absorption and molecular relaxation. Wave characteristics of sound are neglected in acoustic analysis, and sound propagation is assumed to follow acoustic rays. The equation is therefore solved along sonic boom propagation trajectories. The in-house code bBoom^[24] that has been validated against abundant experimental measurements will be used for sonic boom prediction.

Near-field waveforms given by CFD are utilized to validate the far-field propagation approach. The predicted far-field waveform shows good consistency with SBPW data (Figure 4), which verifies high accuracy of the method used in the current work.

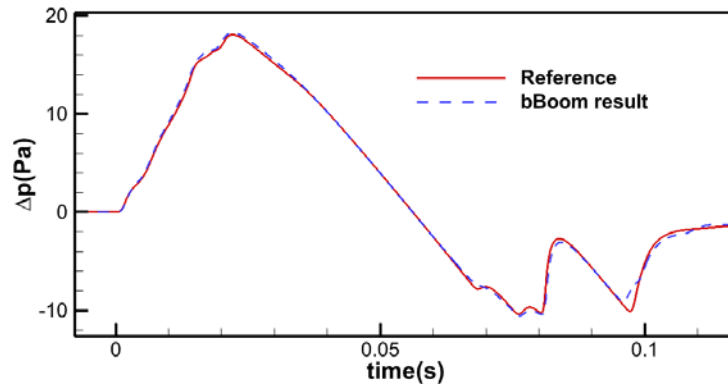


Figure 4 - Comparison of far-field signature of JWB with reference result

3. Baseline SBWB Configuration

3.1 Parametric Method and Baseline Configuration Generation

The TAW fuselage diameter is constrained by cabin size, and its sectional area increases quadratically with the number of seats per row. For large SSTs, this leads to a substantial increase in fuselage volume and raises the difficulty of sonic boom design. In comparison, the BWB adopts a non-circular fuselage that conforms more closely to the cabin, providing a more compact volume distribution and lower volume penalty at the same seating capacity (Figure 5). However, the high integration of the cabin and wing renders the traditional split design framework inapplicable, thereby requiring a full-aircraft parametric method dedicated to the SBWB configuration.

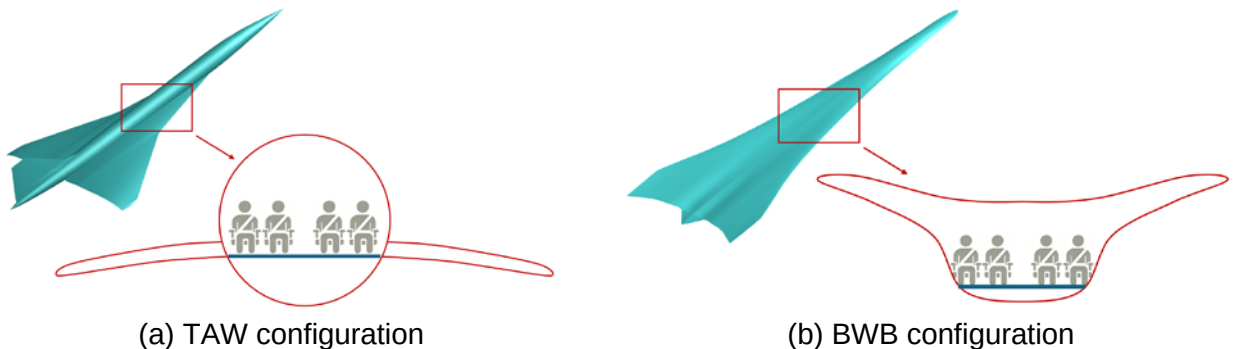


Figure 5 - Comparison between TAW and BWB

Low Boom Design and Analysis of a Blended-wing-body configuration for Large-sized Supersonic Transport

To address this issue, a full-aircraft CST parametric method (Figure 6) is proposed to ensure the global geometric continuity and smoothness of the configuration during optimization. In this approach, the SBWB geometry is defined by a set of closed sectional curves, where all sections are parameterized using CST functions of identical order. Third-order NURBS interpolation is employed between adjacent sections to construct aerodynamic surfaces and maintain second-order geometric continuity. Additional design parameters include the streamwise coordinates of leading/trailing edges as well as the vertical offset of each section. Low-boom design implemented based on this unified parametrization eliminates the need for repeated FFD box reconstruction and re-parameterization induced by geometric deformation.

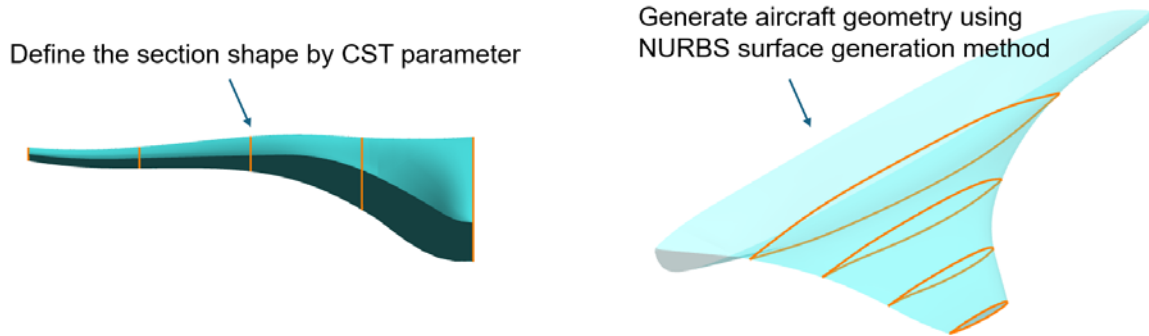


Figure 6 - SBWB parametric method based on CST parameters

The cabin envelope is established as the primary geometric constraint for SBWB parametric modeling. A 160-seat layout with 40 rows and four seats per row is configured, and major cabin dimensions are specified to meet passenger comfort requirements. Due to the limited internal space of supersonic transport, overhead luggage racks are eliminated to prevent spatial crowding. Baggage storage is instead arranged in the tapered section of the rear fuselage. (Figure 7)

The longitudinal symmetry-plane contour is generated according to the cabin boundary dimensions. The overall aircraft length is configured with a reserved height margin to compensate for geometric shrinkage in the spanwise direction. Key control points at the front, rear, upper and lower boundaries of the cabin symmetry plane are selected, and a CST curve is fitted to construct baseline profile. During the baseline configuration design, offsets are applied to the nose and tail to increase effective aircraft length, providing a favorable geometric space for the subsequent low-boom design.

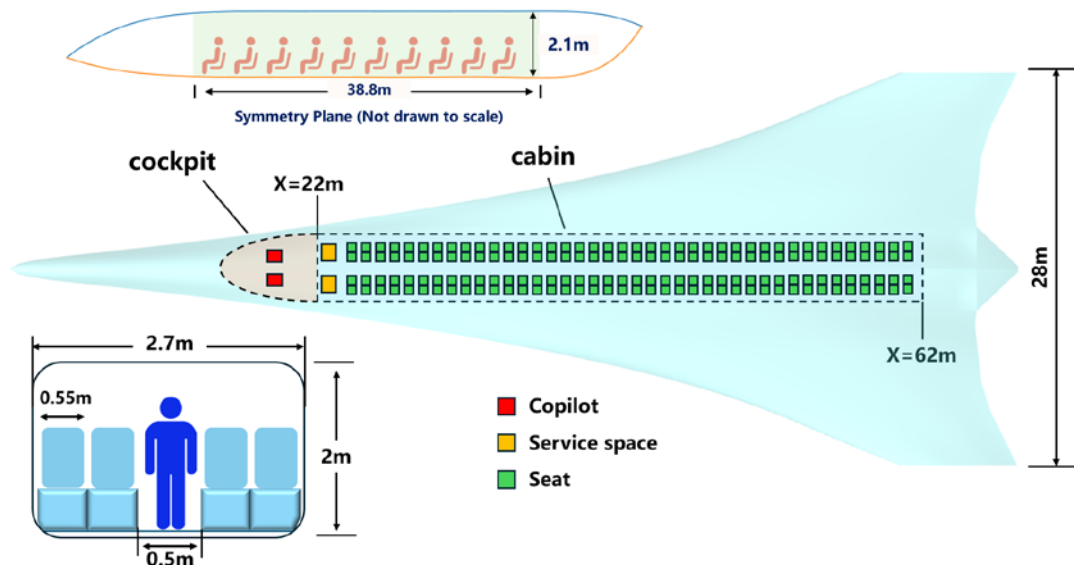


Figure 7 - Demonstration of cabin Interior space

A 3%-thickness thin airfoil is applied uniformly to the outboard wing sections away from the cabin. The sectional profiles of the middle fuselage are acquired by interpolating the airfoil parameters between the outboard wing and the symmetry plane, and the initial aerodynamic configuration is thus constructed.

Considering flight efficiency and sonic boom characteristics, the cruise condition is set as a cruise

Low Boom Design and Analysis of a Blended-wing-body configuration for Large-sized Supersonic Transport altitude of 18 km and a cruise Mach number of 1.8. The takeoff total weight of the SBWB is 170 t, and through a preliminary flight profile analysis, fuel consumption before cruise is about 20 t, resulting in a cruise weight of 150 t. To match the lift of the shape with the design weight, the leading-edge curve is further parameterized by a cubic polynomial, then a rapid aerodynamic optimization is performed and the baseline configuration for the current research is obtained (Figure 8).

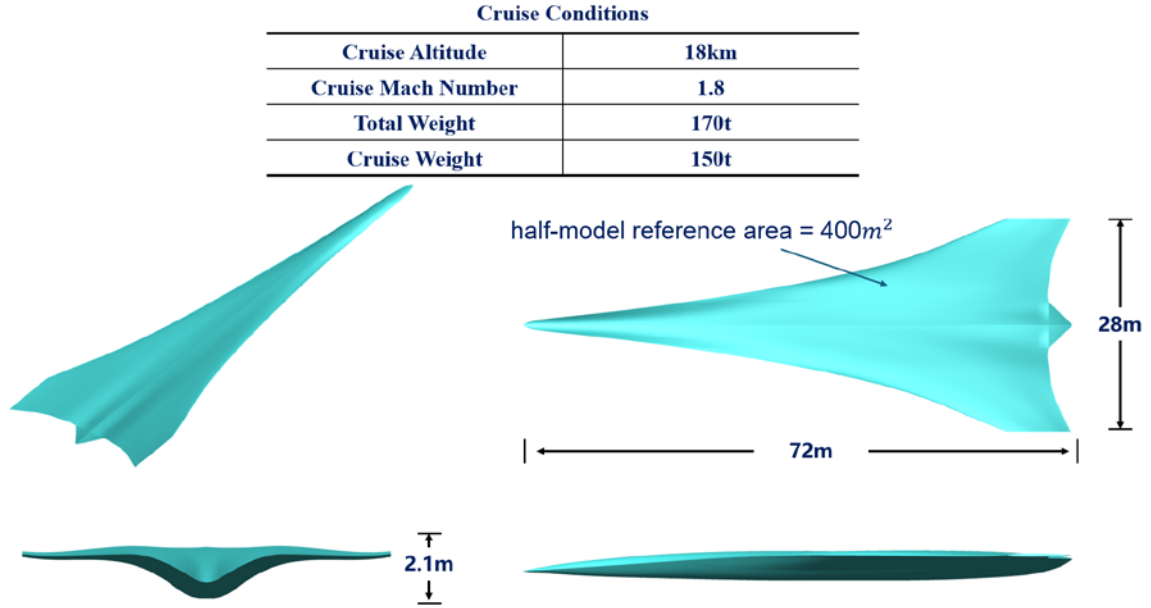


Figure 8 – Sketch of baseline configuration

3.2 Aerodynamic Analysis of the Baseline Configuration

The sonic boom mesh is then generated according to the acquired cruise angle of attack. The cruise angle of attack is obtained via high-fidelity aerodynamic simulation with a relatively coarse aerodynamic mesh, i.e. 4.1 degrees. The cruise lift-to-drag ratio is 11.2. The lift coefficient is 0.107. Pressure drag and viscous drag account for 62.2% and 37.8% respectively, indicating that most of the drag is generated by volume, which is consistent with the basic assumption of large-sized SBWB. As shown in the pressure distribution (Figure 9), the low-pressure region of the SBWB appears earlier than that of the TAW configuration. The TAW configuration is typically designed with a very slender nose, which generates lift approximately from a position near the half-length of the fuselage. However, the non-lifting region of the SBWB is limited to only a portion of the nose. The early generation of lift may provide a new direction for forebody waveform design. In JSGD inverse design for TAW configurations, the nose shock intensity generally must be enhanced, as a stronger bow shock serves to prevent shock coalescence. The earlier lift generation of the SBWB allows part of the equivalent area at the nose to be contributed by aerodynamic lift, thereby reducing drag produced at the nose.

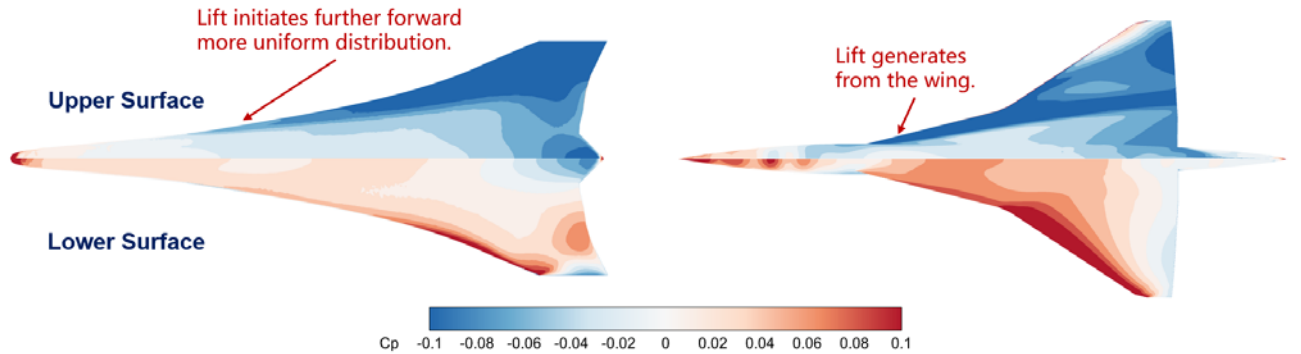


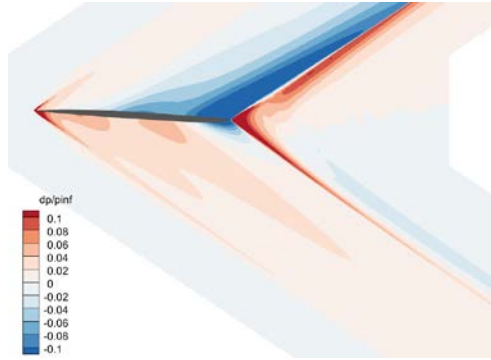
Figure 9 – Pressure distribution on upper and lower surfaces

Figure 10 illustrates the near-field symmetry-plane overpressure distribution derived from sonic boom numerical analysis. The forebody pressure waveform of the proposed SBWB configuration exhibits a generally similar variation trend to that of the conventional TAW configuration. Nevertheless,

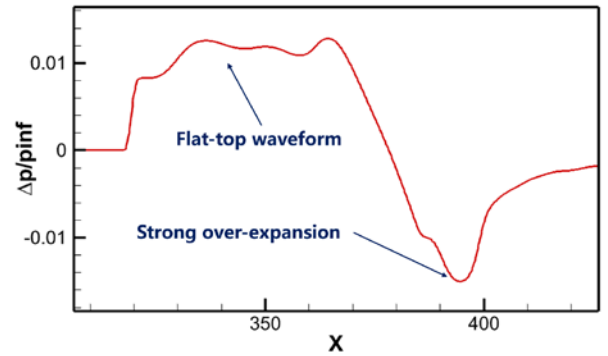
Low Boom Design and Analysis of a Blended-wing-body configuration for Large-sized Supersonic Transport

constrained by the substantially extended cabin length of the large-capacity SBWB configuration, the tapered transitional section of the rear fuselage is severely compressed in the longitudinal direction. The abrupt volumetric contraction of the aft fuselage induces intense local over-expansion flow, which constitutes a prominent unfavorable factor for low-boom design. In this regard, compared with conventional TAW supersonic schemes, the SBWB configuration exhibits more pronounced afterbody waveform defects, indicating that subsequent low-boom optimization for the SBWB should focus primarily on the suppression and regulation of adverse rear-body wave characteristics.

Furthermore, the sonic boom signal extracted at a downstream distance of three times the fuselage length ($R/L = 3$) beneath the aircraft is analyzed in detail. For comparative reference, the near-field waveform of conventionally designed configurations without low-boom optimization typically contains two intense shock waves, which are separately induced by the volumetric discontinuity at the nose and the lift discontinuity at the wing leading edge. In contrast, the fore-body waveform of the baseline SBWB in this study exhibits a flat-top profile. This waveform characteristic is attributed to its extended lift distribution range and the axial coupling effect between volumetric variation and lift variation along the fuselage. In the traditional low-boom design of TAW configurations, the classic split design concept assumes that the modification of fuselage volume distribution exerts negligible influence on aircraft lift performance. However, the analytical results indicate that this conventional assumption is no longer applicable to the SBWB configuration, which further demonstrates the necessity of implementing integrated aerodynamic and low-boom design for the SBWB.



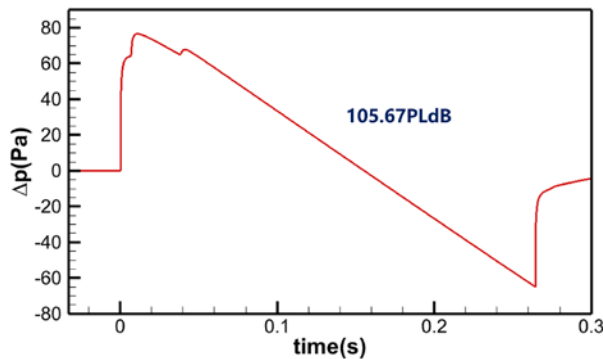
(a) Overpressure distribution



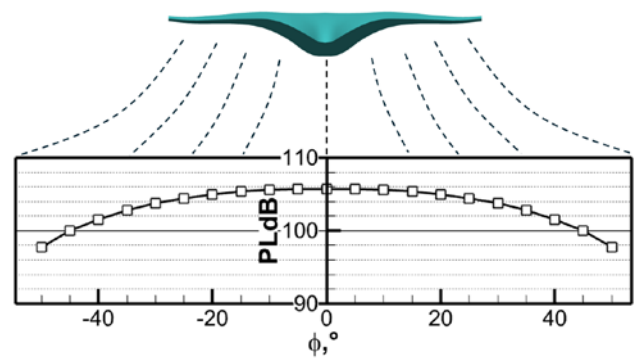
(b) $\Delta p/p_\infty$ at $R/L = 3$

Figure 10 – Overpressure distribution and sonic boom signature at $R/L = 3$

Far-field propagation simulation based on near-field sonic boom waveforms reveals that the ground sonic boom intensity directly beneath the aircraft reaches 105.67 PLdB (Figure 11), a magnitude comparable to that of the Concorde airliner. Observations of the far-field waveform indicate that although its overall trend is close to a typical N-wave, the leading shock waves are not fully merged and display indistinct multiple shock wave characteristics. This phenomenon demonstrates that the overall sonic boom intensity of the SBWB configuration is primarily determined by shock wave merging and the large amplitude induced by high lift. Accordingly, the classic JSJD theory is separately applied to conduct low-boom design on the SBWB model, to explore the effective potential of sonic boom suppression for this configuration.



(a) Ground waveforms



(b) carpet PLdB of SBWB

Figure 11 – Ground waveforms and carpet PLdB distribution

4. Result and Analysis

4.1 Fore Body Waveform Analysis and Inverse Design

An effective length of 68 meters is adopted in this paper to generate the JSGD equivalent-area distribution target. Figure 12 shows the target F-function and a comparison between the equivalent-area distribution of the baseline configuration and the target equivalent-area distribution adopted in this paper.

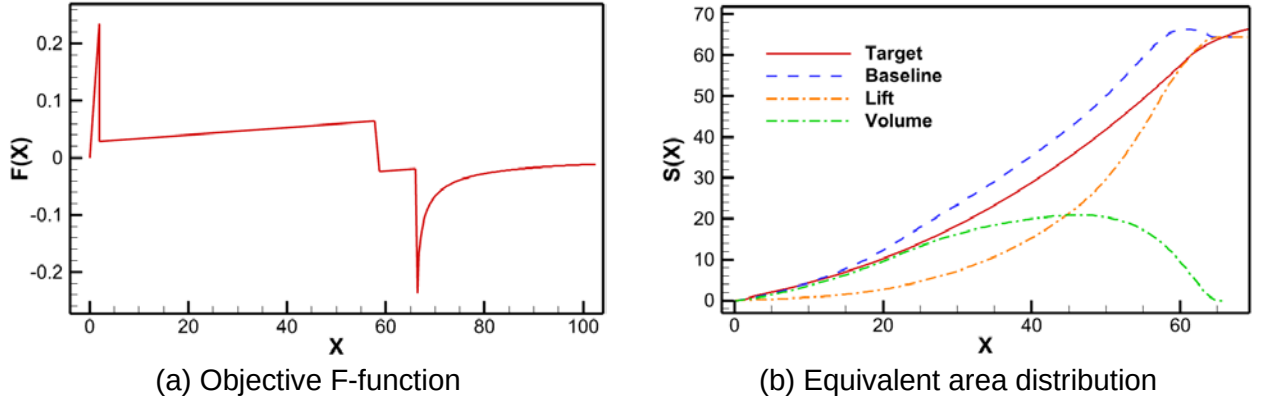


Figure 12 - Objective F-function and equivalent area distribution

It is evident from the equivalent-area distribution that the large-volume cabin governs the overall equivalent-area variation. The 160-seat cabin structure is analyzed and modeled independently, with conical profiles adopted to approximate the nose and tail sections to keep the start and end positions of the equivalent-area distribution consistent with those of the full aircraft configuration. Since practical shapes must maintain convex characteristics, this profile serves as the lower limit of equivalent area under the designated cabin volume. The geometric boundary of minimum equivalent area is then enclosed by circular fuselage sections, and the relevant sectional area distribution is compared with that of the baseline SBWB configuration as illustrated in Figure 13. The comparison indicates that the equivalent area of circular cabin sections is substantially larger than that of the SBWB structure. This demonstrates that circular fuselage forms impose more stringent constraints on equivalent-area adjustment in large-sized SST design, which increases design difficulty and may even prevent the target distribution from being achieved.

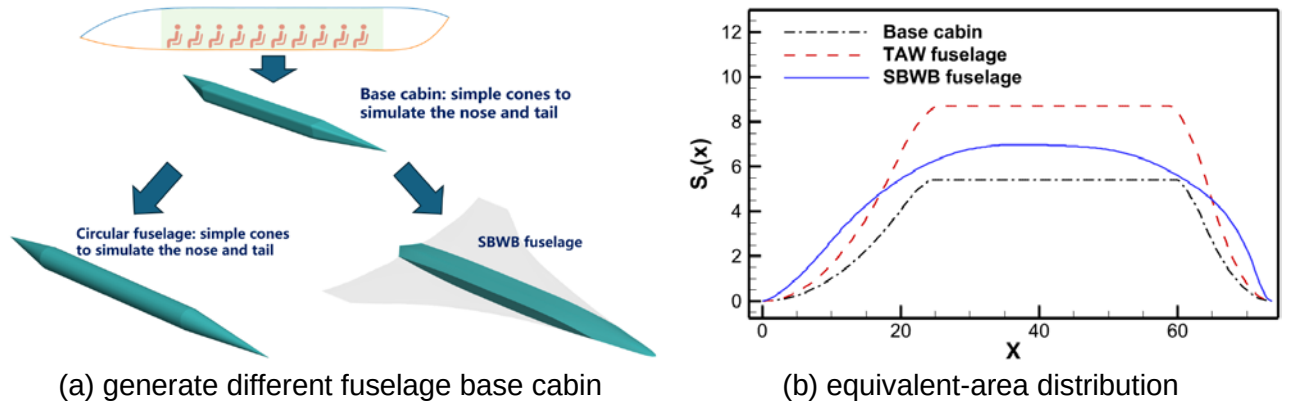


Figure 13 – Fuselage-area distribution of TAW and SBWB

JSGD inverse design is conducted in this study using the full-aircraft parameterization method presented in Section 3.1. Ten sections are defined with fifth-order CST parameters. The spanwise stations of all sections are kept fixed and excluded from design variables.

To address slow convergence in high-dimensional optimization, the design is divided into two stages. The first stage performs planform optimization with the CST parameters of each section fixed. The leading/trailing edge positions, z-direction offsets and relative trailing edge offsets of individual sections are optimized, while parameters at the symmetry plane remain unchanged. With 37 design variables, this stage minimizes the integral of deviations between actual and target equivalent-area distributions. It barely changes the aircraft volume and mainly aims to obtain an optimal lift distribution. The second stage refines section shapes with the planform fixed, involving 120 design variables. A

block-based strategy is adopted: random subsets are selected to update the target equivalent-area without flow computation, and the differential evolution algorithm is employed for sub-optimization. Only results that improve the equivalent-area are retained. The 120 variables are split into 10 groups; after three regrouping cycles, 20 of 30 iterations are accepted. Figure 14 presents the optimized equivalent-area distribution and the near-field waveform obtained from the rapid sonic boom prediction method.

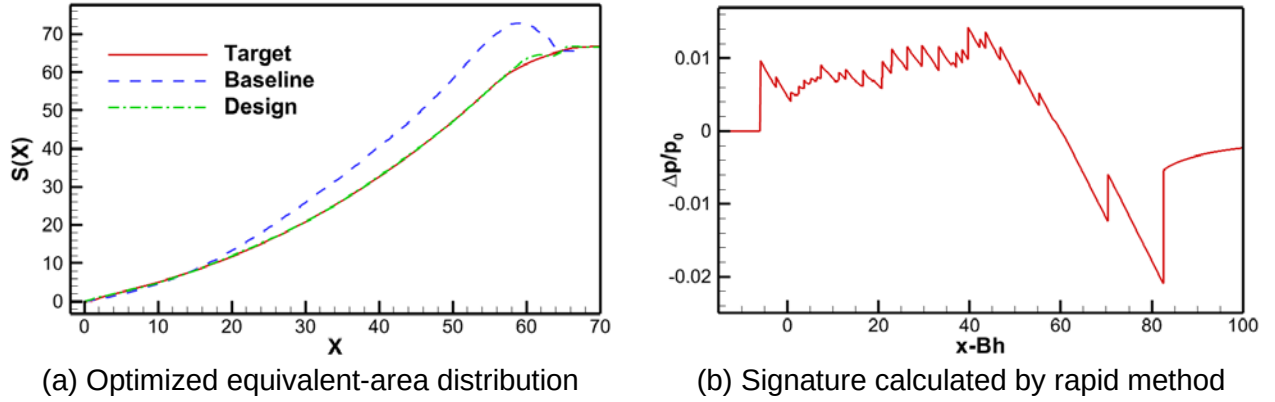


Figure 14 - Optimized equivalent area distribution and near-field overpressure signature

The optimized configuration is evaluated for aerodynamic and sonic boom characteristics using the high-fidelity CFD method (Figure 15). Due to inherent discrepancies in the lift analysis performed by the panel method, the cruise angle of attack of the inverse-designed configuration under high-fidelity analysis deviates from the panel method result, and consequently the calculated sonic boom signature does not fully match the JSGD target waveform. It is observed that the optimized rear-body waveform generates a relatively strong shock wave. This is primarily attributed to insufficient equivalent-area matching at the rear-body section. The rapid contraction of the empennage intensifies expansion waves, which to a certain extent inhibits the forward merging of shock waves formed at the fuselage trailing edge.

It is worth noting that the proposed configuration shares a similar takeoff weight with the Tu-144, while offering a larger passenger capacity. Qiao et al.^[1] implemented JSGD inverse design on the Tu-144 platform but failed to attain the ideal ramp-shaped target waveform; It is concluded that constraints of high-altitude flight and large takeoff weight create major barriers to low-boom shaping. By comparison, the SBWB in this study delivers a waveform much closer to the target profile after inverse optimization. This outcome reveals that the SBWB configuration carries inherent advantages for low-boom design of large-scale supersonic commercial airliners.

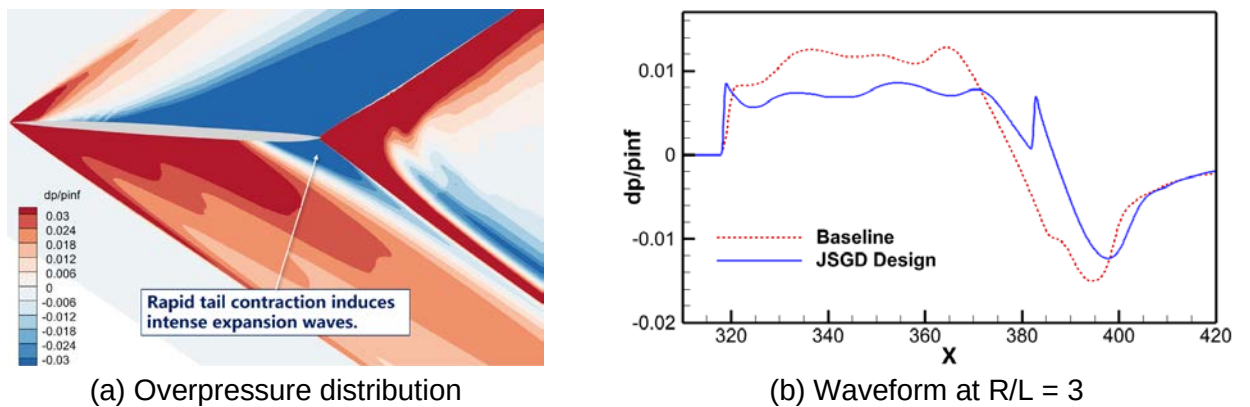


Figure 15 – Optimized overpressure distribution and waveforms at $R/L = 3$

The far-field propagation simulation is performed on the sonic boom signal at three times the fuselage length to obtain the ground waveform and the PLdB distribution at circumferential angles. The sonic boom intensity directly below is reduced to 99.21 PLdB (Figure 16), which is 6.46 PLdB lower than that of the baseline configuration. It is worth noting that the sonic boom reduction effect is more pronounced at positions with small azimuth angles than directly beneath the aircraft. This arises because the overall volume is reduced to a certain extent during optimization, and the volume contribution from lateral positions gradually becomes dominant. Consequently, the boom mitigation

Low Boom Design and Analysis of a Blended-wing-body configuration for Large-sized Supersonic Transport
performance is less constrained by lift requirements.

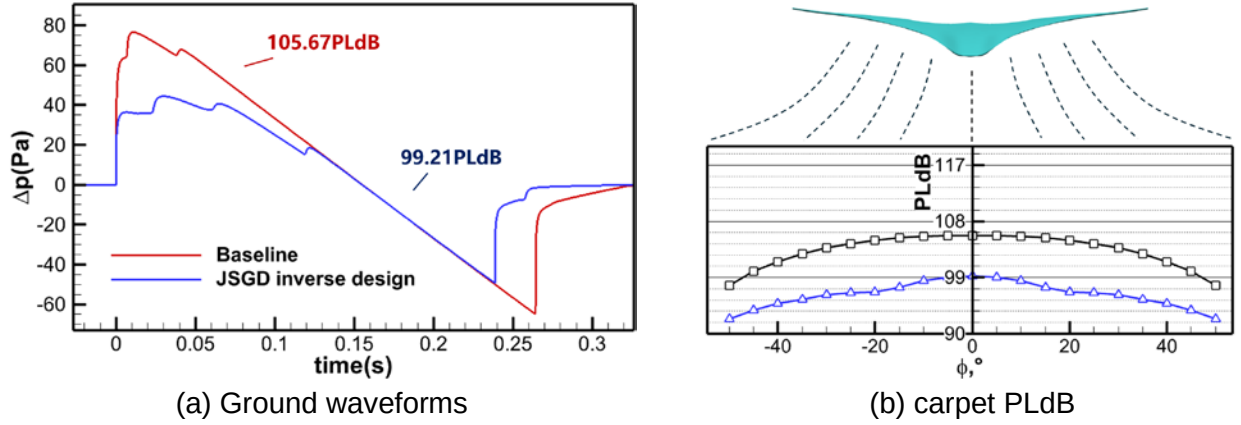


Figure 16 – Optimized ground waveforms and carpet PLdB of SBWB

In this section, the JSGD inverse design method is implemented for low-boom shaping of the SBWB, and the optimized near-field waveforms closely match the theoretical minimum-boom target profile. When benchmarked against equal-weight tube-and-wing supersonic aircraft like the Tu-144, the SBWB achieves superior waveform alignment under the same inverse design workflow. The results verify that classic low-boom design approaches are fully compatible with SBWB geometries, and further underscore the low-boom advantages of SBWB for large supersonic civil transports.

Even after effective forebody waveform optimization, severe over-expansion waves still develop across the rear fuselage of this tailless baseline configuration, forming a clear barrier to deeper sonic boom reduction. Auxiliary aerodynamic components are widely proven as a mature tactic to adjust rear-body pressure waveforms for TAW low-boom aircraft. The subsequent section accordingly investigates the waveform-modulating performance of aerodynamic components when deployed on the SBWB's rear section.

4.2 Rear Body Waveform Analysis

The preceding low-boom design is implemented based on a tailless clean configuration. Although such tailless configurations have emerged as a major research direction, especially in the design of hypersonic aircraft and supersonic fighters, they are generally regarded as unsuitable for civil airliners due to challenging control characteristics. Therefore, it is necessary to incorporate additional components into the basic configuration to account for control performance. In addition, the analysis of sonic boom characteristics of the clean configuration reveals exceptionally strong expansion waves at the rear body, which cannot be effectively mitigated in subsequent low-boom design processes. In the low-boom design of TAW configurations, control surfaces are widely adopted as an effective means of adjusting the rear-body waveform. Whether this method can also be applied to the low-boom design of the SBWB merits further investigation. Therefore, this work evaluates how two standard BWB appendages—winglets and V-tails—alter sonic boom pressure signatures.

The dimensions of winglets and V-tails are calculated using empirical formulas. The planform sizes and installation positions of the two components are shown in Figure 17. Symmetric thin airfoils are adopted for V-tails to balance control efficiency and aerodynamic drag reduction, while winglets employ the same airfoil as the wing tip.

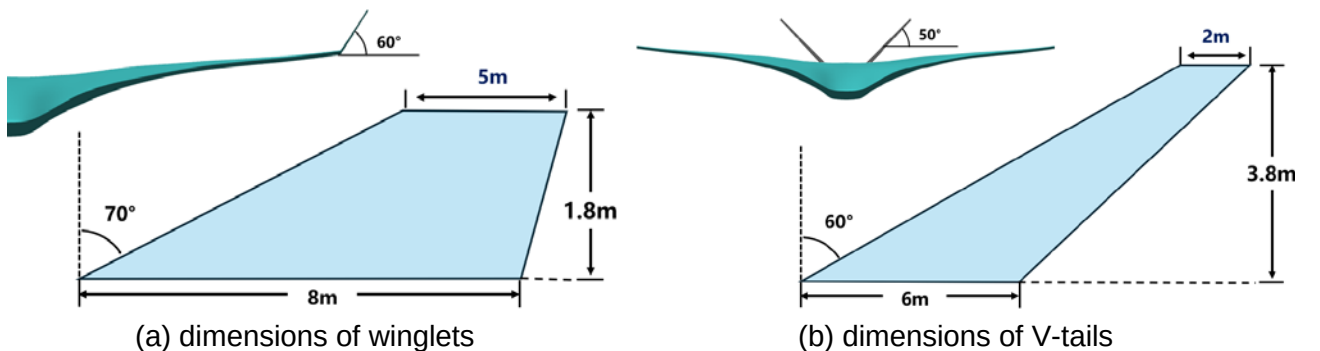


Figure 17 – Geometry of the winglet and V-tail

4.2.1 Winglet effects on sonic boom waveform

It is observed that installing winglets alone slightly intensifies the shock wave formed at the terminal section of the volumetric distribution and alleviates the over-expansion phenomenon of the rear-body to a certain degree. Previous research has confirmed that wingtip vortices propagate downstream along the Mach cone and interact with the over-expansion flow field behind the rear-body shock. Such mutual interaction further amplifies the magnitude of over-expansion through wave superposition. On the one hand, additional flow disturbances induced by winglets superimpose onto the rear-body waveform, leading to a moderate rise in shock wave amplitude. On the other hand, winglets are effective in restraining intense wingtip flow disturbances and hinder the backward propagation of low-pressure flow over the wing upper surface, which prevents these flow features from interacting with the rear-body over-expansion region. Despite the above improvements, the overall suppression effect on over-expansion still does not meet the anticipated design targets.

Further comparative analysis of the rear-body waveform of the clean configuration and the winglet-fitted model indicates that the shock wave generated at the fuselage tail is exceptionally strong. During downstream propagation, the majority of low-pressure disturbances originating from wingtip flow are intercepted by this robust rear-body shock and therefore cannot travel to the symmetry plane to form superimposed flow structures. This result (Figure 18) demonstrates that wingtip vortex disturbances constitute only a small proportion of the factors triggering rear-body over-expansion, even for the original clean configuration. Accordingly, merely suppressing wingtip flow disturbances is insufficient to achieve desirable performance in sonic boom mitigation.

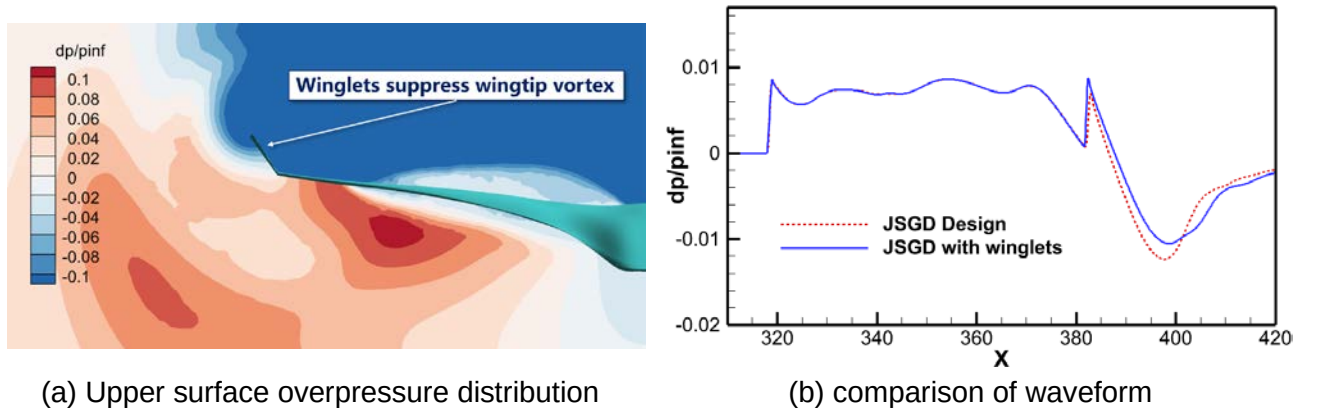


Figure 18 – Influence of winglets on sonic boom characteristics

4.2.2 V-tail effects on sonic boom waveform

In contrast, equipping the aircraft with V-tails alone exerts a negligible influence on the waveform. (Figure 19) Flow field analysis indicates that the symmetric V-tails with zero incidence angle generate minor flow disturbances, most of which are shielded by the fuselage and cannot affect the sonic boom waveform below, resulting in limited capability for wave system regulation. It is worth noting that V-tails with preset incidence angles are adopted in some existing designs to compensate for lift. As anticipated, such configurations produce stronger flow disturbances and may exert a more pronounced impact on the waveform.

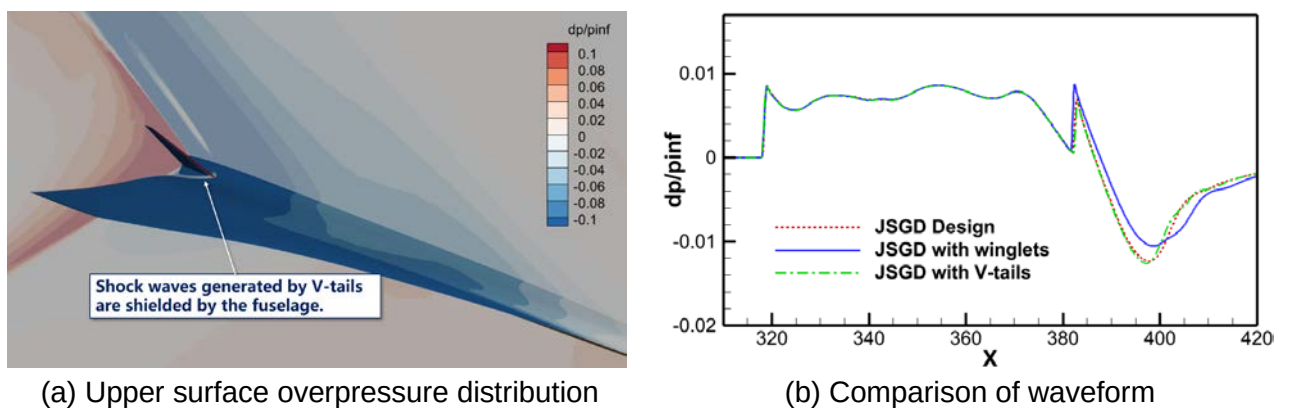


Figure 19 – Influence of V-tails on sonic boom characteristics

Subsequently, winglets and V-tails are both mounted on the JSGD-designed configuration for sonic boom characteristic analysis. According to the near-field waveform results (Figure 20(a)), although V-tails exert negligible influence on the wave system when applied alone, the combined effect of the two components differs from the simple superposition of their individual contributions after integrated installation. The rear-body over-expansion intensity generated by the coupling effect of the two components is further marginally reduced, and the shock wave enhancement induced by winglets is also suppressed to a certain extent. This indicates that a mutual interference effect exists between the flow disturbances produced by winglets and V-tails, which improves the near-field wave system structure and achieves a partial reduction of the adverse aerodynamic wave characteristics. Near-field waveform signals at different azimuthal angles are propagated to the ground for far-field sonic boom evaluation (Figure 20(b)). The results demonstrate that the ground sonic boom intensity directly beneath the aircraft is reduced from 99.2 PLdB to 98.7 PLdB after the combined installation of winglets and V-tails, confirming the effective sonic boom suppression capability of the compound aerodynamic configuration.

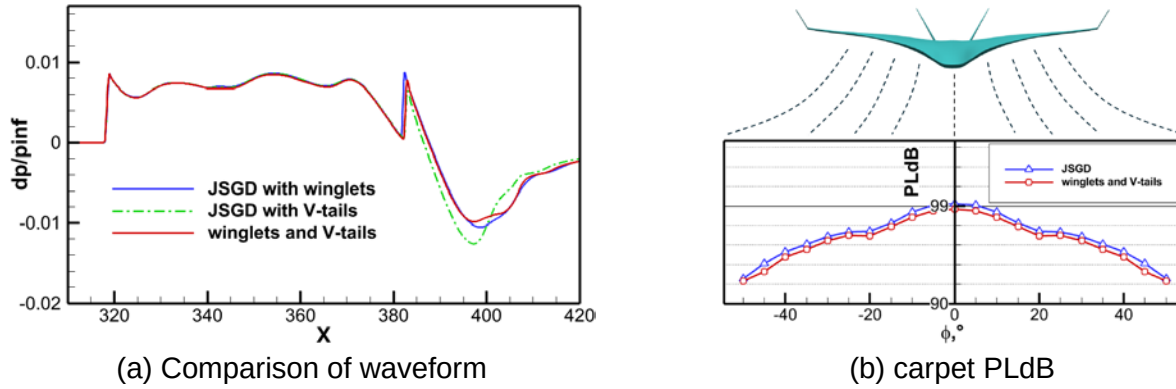


Figure 20 - Influences of aerodynamic components on sonic boom characteristics

The influences exerted by winglets and V-tails on the sonic boom characteristics of the SBWB configuration have been assessed in the present work. Rear-body over-expansion can be effectively mitigated with the installation of winglets. Only marginal waveform tuning improvement can be attained when V-tails are applied independently, while strengthened regulation performance can be realized when V-tails are coupled with winglets. It should be noted that severe rear-body over-expansion of the adopted configuration may partially obscure the tuning effects of these aerodynamic fittings. Supplementary investigations need to be conducted to clarify the fundamental mechanisms through which aerodynamic appendages modify the sonic boom performance of the SBWB.

5. Conclusion

The aerodynamic feasibility and low-boom feasibility of a 160-seat large-sized SBWB are examined in this paper. It is found that the SBWB configuration is characterized by a smaller cruise angle of attack and a wider lift distribution range. Moreover, its fuselage conforms more closely to the cabin structure, which facilitates equivalent-area distribution matching in low-boom design. Based on the preceding research results, the classic JSGD inverse design method is subsequently adopted to conduct low-boom design for the SBWB, and the ground sonic boom intensity is reduced from 105.67 PLdB to 98.7 PLdB. The rear-body waveform of the SBWB is then analyzed; the results indicate that winglets can shield the expansion on the upper wing surface, which contributes to suppressing sonic boom levels, while most of the impact of V-tails on the sonic boom waveform is blocked by the rear fuselage. It is acknowledged that V-tails exert a notable influence on pressure waveforms, and rational design of V-tails can effectively achieve the objective of wave system regulation. The research results demonstrate that large-sized SST possesses economic advantages and can achieve sonic boom mitigation through low-boom design.

Future research may further adopt methods to design the fore-body waveform, and canards may be introduced for waveform control if necessary. For the rear-body, the over-expansion problem must be addressed first, and the research focus should remain on weakening the shock waves induced by volume termination, thereby reducing the intensity of over-expansion. The findings of this paper indicate that the sonic boom characteristics of the SBWB configuration differ substantially from those of the TAW configuration. In addition to further research on large-sized SBWB, the application of the SBWB configuration to the design and research of supersonic business jets may also be explored.

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